

High-Performance Parallel MPI Grid Decomposition Guide

Technical Blueprint on 2D Spatial Domain Splitting, Halo Exchanges, and NetCDF4 Parallel I/O on National Supercomputers (Pratyush & Mihir)

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1. Executive Overview & Mathematical Formulation

In atmospheric general circulation and non-hydrostatic Numerical Weather Prediction (NWP) models, simulating the Navier-Stokes fluid equations over the Indian subcontinent at 1.2 km resolution requires decomposing continuous 3D domain grids into partitioned sub-domains managed by distinct MPI ranks across multi-node supercomputing clusters.

CFL Stability Criterion: For explicit time-stepping discretization, the Courant–Friedrichs–Lewy parameter must strictly satisfy: $C = u \cdot (\Delta t / \Delta x) \leq C_{max} \leq 1.0$. When grid resolution Δx is refined from 12 km to 1.2 km, the time-step Δt must be reduced ten-fold, multiplying inter-node communication intensity.

2. 2D Domain Decomposition Topology & Halo Exchanges

Modern dynamical cores utilize 2D Cartesian process grids ($nproc_x \times nproc_y$). Each process owns an interior subdomain of dimensions ($n_x \times n_y \times n_z$) surrounded by ghost cell buffer layers (halo zones) of depth 2 to 4 grid points to compute horizontal spatial derivatives without boundary data stalls.

Decomposition Mode	Scalability Limit	Memory Overhead	Operational Recommendation
1D Slab (Zonal Splitting)	< 512 MPI Ranks	High (Full Latitudinal Rows)	Diagnostic offline post-processing only
2D Block Cartesian	10,000+ Cores (Pratyush)	Optimal (~2-4% Halo Overhead)	Standard operational baseline for NCUM / WRF
Octree Adaptive Mesh (AMR)	Dynamic Extreme Scale	Variable Cache Profile	Experimental high-resolution cyclone tracking

2.1 Non-Blocking Boundary Exchange Protocol

```
! Fortran 2008 / OpenMPI 4.1 Non-blocking Halo Exchange
CALL MPI_Irecv(halo_east, nz*ny, MPI_REAL8, nbr_east, 101, MPI_COMM_WORLD, req(1), ierr)
CALL MPI_Isend(send_west, nz*ny, MPI_REAL8, nbr_west, 101, MPI_COMM_WORLD, req(2), ierr)
! Interleave interior grid calculations to hide network latency
CALL compute_internal_divergence(p_field, u_field, v_field)
CALL MPI_Waitall(2, req, MPI_STATUSES_IGNORE, ierr)
```

3. Parallel I/O Optimization with NetCDF4 & HDF5

At sub-kilometer grid scales, disk write bottlenecks during model checkpointing can consume 40% of runtime if ranks write sequentially. Utilizing collective MPI-IO with POSIX file striping on Lustre partitions guarantees sustained throughput > 120 GB/s.

Operational Best Practice: Always configure Lustre striping count across OST partitions matching the number of writer aggregator nodes: `lfs setstripe -c 32 -S 4M /scratch/nwp_forecast/`.